
Refinement Buys Intelligibility, Search Buys Identity: What Test-Time Compute Buys in Masked-Diffusion TTS

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 tensorViz

Code: <https://github.com/nityanandmathur/aspect-d>

Models: <https://huggingface.co/nityanandmathur/aspect-d-masked-diffusion-tts>

Abstract

We ask what more refinement steps and search improve in masked-diffusion text-to-speech (TTS). We train 15 width–depth shapes (3 seeds each, 19–133 M non-embedding parameters) on 2,000 h of English speech and vary the refinement steps T only at inference, testing on 400 zero-shot items from 174 held-out speakers. From unintelligible one-step output ($T=1$, word error rate (WER) 113.8%) to the default $T=16$, refinement closes 86.2% of the WER gap to real recordings but only 46.4% of the gap in speaker similarity, a ratio of $1.86\times$. On 9 retrained runs the ratio shrinks with training ($1.84\times$, $1.36\times$ and $1.23\times$ at 30k, 90k and 180k steps) and stays above one at all three lengths. At the same 128 generator passes, keeping the better of two samples, as judged by one speaker encoder, beats one $T=16$ sample by +0.0153 in similarity under another encoder and wins on 57.4% of items; with the encoders swapped the gain is +0.0089, against +0.0827 from $3\times$ longer training. Search costs +0.0408 absolute WER and 7.1–7.9% extra compute. The codec loses 62% of the similarity gap between our best 90k-step model and real audio. We claim only that refinement helps intelligibility more than speaker identity, for small models, one English corpus, one codec and automatic metrics.

1 Introduction

A diffusion language model can spend serial compute in two ways: through its depth d , fixed in training and paid on every forward pass, or through its refinement steps T , chosen per request at inference. A standard autoregressive transformer has only the first. If extra steps could do the work of extra layers, a smaller model could be served with more steps, but only for the qualities that more steps improve.

Zero-shot text-to-speech (TTS) has two capabilities that need not improve at the same rate: intelligibility, measured by word error rate (WER), and speaker identity, measured by speaker similarity to the original prompt (SIM-o). We train one grid of masked-diffusion codec TTS models [Borsos et al., 2023b, Wang et al., 2025b]. In each of three parameter budgets, the shapes split a fixed parameter count differently between width and depth, with 3 seeds each. We vary T only at inference and compare each metric with its value on real recordings (§3).

Refinement mostly fixes the words. From $T=1$ to the default $T=16$ it closes 86.2% of the reachable WER range and 46.4% of the reachable SIM-o range, a ratio of $1.86\times$ (§4.1). At $T=1$ the output is unintelligible, and the ordering also holds when counting starts at $T=2$ ($2.41\times$) or $T=4$ ($3.25\times$). On 9 retrained runs the ratio shrinks with training but stays above one: $1.84\times$, $1.36\times$ and $1.23\times$ at 30k, 90k and 180k steps (§4.2).

Search does raise identity. With the same 128 generator forward passes (matched NFE), best-of-2 search beats $T=16$ in speaker similarity by +0.0153 when one encoder (WavLM-L) selects and another (ECAPA) scores, and by +0.0089 with the roles swapped (Table 3). It costs +0.0408 absolute WER and 7.1–7.9% extra compute for selection.

Before training, we pre-registered four hypotheses about a power law fitted to the error as a function of width, depth and T , two of them declared primary (App. B). The first primary hypothesis, that WER falls faster with T than identity error, passed its registered test, but the tested quantity changes sign when the error is rescaled, for example by a square root (§4.5). The second, a fixed exchange rate between depth and steps, failed its registered criterion, and the exponent that would set it sits at the limit set for the fit, so we could not identify an exchange rate. The shape hypothesis could not be decided, because the coefficient of the width term also hits its limit, and the prediction from small to large budgets failed. Our main measurement, the share of the reachable range, was added after the registered analysis and is not a pre-registered test.

Contributions. ❶ A seeded grid of 45 masked-diffusion TTS models with T varied at inference (75 points per metric), released with code and run records (App. A). ❷ A measure that needs no fitted model: for each metric, the share of the distance to real audio that refinement closes, checked from later starting steps, without failed outputs and after longer training. ❸ A comparison, in one unit (SIM-o under WavLM-L), of what raises speaker identity beyond the default $T=16$: longer training raises it most, then more parameters (Table 4). At the same NFE, search gains about twice as much as speaker-contrastive guidance (+0.0195 against +0.0087 at NFE 256), and its gain grows with more candidates.

These results come from one small regime. Models have 19–133 M non-embedding parameters, plus one 276 M budget, and train for 30k steps on 2,000 h of English from one corpus [He et al., 2024]; the 90k and 180k runs repeat the same data. We use one codec, one sampler, no classifier-free guidance in the main grid, and automatic metrics only. Our best 90k-step model reaches SIM-o 0.4807, below the SIM-o ceiling (0.5554) and below F5-TTS [Chen et al., 2025b] on the same items (0.6549). We claim the ordering of the two shares within this regime only.

2 Related work

Masked diffusion for language and speech. Discrete diffusion with an absorbing mask state [Austin et al., 2021] was later simplified and improved [Lou et al., 2024, Sahoo et al., 2024, Shi et al., 2024], given a scaling law [Nie et al., 2025a] and trained at 8B parameters [Nie et al., 2025b]. These studies scale training compute and treat sampling steps as a trade-off between speed and quality. SoundStorm [Borsos et al., 2023b] brought MaskGIT confidence decoding [Chang et al., 2022] to residual codebooks, decoded one level at a time as in our sampler. MaskGCT [Wang et al., 2025b], NaturalSpeech 3 [Ju et al., 2024] and recent masked or block diffusion models [Fan et al., 2026, Madha et al., 2026] also generate discrete speech tokens by masked prediction. Flow matching [Le et al., 2023, Eskimez et al., 2024, Chen et al., 2025b] is the main continuous alternative, and autoregressive models [Borsos et al., 2023a, Chen et al., 2025a, Anastassiou et al., 2024, Du et al., 2024, Ye et al., 2025] emit tokens left to right; their token stage has no refinement steps, although Seed-TTS and CosyVoice render the tokens with a diffusion or flow-matching decoder. Step ablations exist. SoundStorm finds that audio quality stops improving past its default number of iterations, F5-TTS reports WER and SIM-o at two NFE settings, and MaskGCT notes that WER improves far more than SIM-o over its step sweep. Each covers a few operating points of one model, and raw changes in WER and SIM-o have no common unit. We measure both as the share of the distance to real audio that refinement closes, across many shapes at equal parameter count.

Test-time compute: refinement and search. Language models often spend test-time compute by sampling candidates and keeping the one a verifier prefers [Cobbe et al., 2021]. At matched FLOPs this can replace a much larger model on problems the small model sometimes solves, but not on the hardest ones [Snell et al., 2025]. In continuous diffusion, gains from more denoising steps flatten after a few dozen steps, while verifier-guided search over the sampling noise keeps improving samples [Ma et al., 2025]. Masked diffusion commits tokens for good. ReMDM adds remasking so that extra steps keep helping [Wang et al., 2025a], and UnMaskFork searches over unmasking trajectories [Misaki and Akiba, 2026]. In TTS, Llasa finds that beam search on a speaker-similarity verifier raises SIM but can leave WER worse than without search, unless a WER verifier is added

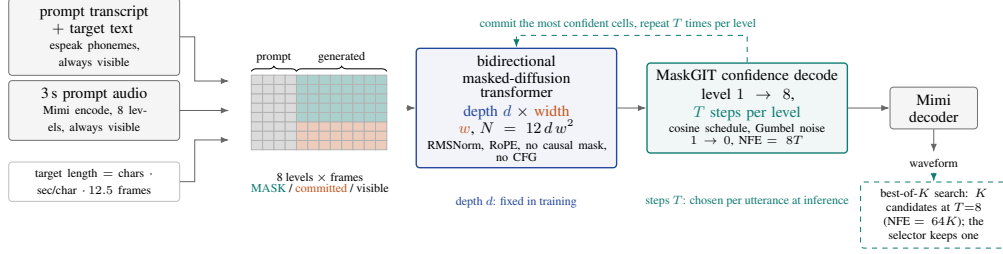


Figure 1: *Generation pipeline*. Phonemes and a 3 s prompt stay visible while the masked target is decoded. Depth d is fixed in training; T (NFE = $8T$) and best-of- K search are set at inference.

[Ye et al., 2025], and best-of- N gains depend on whether the verifier and the evaluating ASR share a model family [Yu and Kang, 2026]. We compare refinement and search in one sampler without remasking, at equal NFE, and test whether the search gain depends on the selecting and scoring speaker encoders sharing an architecture (§4.3).

Depth, width and iterated depth. At a fixed non-embedding parameter count, loss depends only weakly on transformer shape [Kaplan et al., 2020], and compute-optimal scaling is stated in parameters and tokens [Hoffmann et al., 2022]. Yet depth beats width only below a width-dependent threshold [Levine et al., 2020], downstream quality depends on shape when pre-training loss does not [Tay et al., 2022], and deeper models of equal size generalise more compositionally, with diminishing returns [Petty et al., 2024]. Weight-tied recurrence trades parameters for iterations [Dehghani et al., 2019, Saunshi et al., 2025, Geiping et al., 2025], and in masked diffusion, recursive depth can partly replace denoising steps [Carballo-Castro et al., 2026]. In our models, layers do not share weights and each refinement step fixes some tokens for good, so a step is not simply another pass through the same layers. We could not identify a fixed exchange rate between depth and steps (§4.5).

3 Method

Model and training. The generator is a bidirectional pre-LN transformer [Xiong et al., 2020] over the 8 coarsest residual codebooks of the Mimi codec [Défossez et al., 2024] at 12.5 Hz. A model has width w , depth d , head dimension 64, a GELU MLP of ratio 4 [Hendrycks and Gimpel, 2016], RMSNorm [Zhang and Sennrich, 2019], RoPE [Su et al., 2024] and no weight tying, so its non-embedding parameter count is $N = 12 d w^2$ [Kaplan et al., 2020]. Training uses the coarse-to-fine masking of SoundStorm [Borsos et al., 2023b]: each example masks part of one random level and all finer levels, and the loss is cross-entropy on the masked cells of that level. There is no separate speaker encoder and no classifier-free guidance (CFG) [Ho and Salimans, 2022]. The target length is the number of characters in the target text times 0.0602 s, the median duration per character of training clips.

Refinement steps and search. Inference uses MaskGIT confidence decoding [Chang et al., 2022], one codebook level at a time from coarse to fine, with the same number of *refinement steps* T per level. A sample therefore costs NFE = $8T$, where NFE is the number of generator forward passes per utterance. Each step predicts all masked cells, samples tokens at temperature 1 and commits the most confident ones; committed cells are never masked again. The cosine schedule, the Gumbel confidence noise and its seeds are fixed across models and T . The *default* is $T=16$. *Best-of- K search* draws K candidates at $T=8$ each (NFE = $64K$), and a *selector* keeps the candidate whose speaker embedding is closest to the prompt. At *matched NFE* it is compared with refinement at $T=8K$: NFE 128, 256 and 512 for $K=2, 4, 8$. Its *selection cost* is the compute it adds on top of the generator: one Mimi decode and one selector embedding per candidate.

Metrics and reference points. Intelligibility is WER from the automatic speech recognition (ASR) model Whisper-large-v3 [Radford et al., 2023] with greedy decoding, checked by a wav2vec 2.0 CTC recogniser [Baevski et al., 2020, Xu et al., 2021]. Speaker identity is SIM-o [Le et al., 2023], the cosine similarity of speaker embeddings of the output and the original prompt recording. Its encoder, *WavLM-L*, is the seed-tts-eval verification model [Anastassiou et al., 2024]: WavLM-large [Chen et al., 2022] with an ECAPA-TDNN-small head [Desplanques et al., 2020].

Because WavLM-L also picks the search candidate, we score search with a second encoder, *ECAPA* (the SpeechBrain [Ravanelli et al., 2021] ECAPA-TDNN [Desplanques et al., 2020] trained on Vox-Celeb [Nagrani et al., 2017, Chung et al., 2018]). In a swapped control, ECAPA selects and WavLM-L scores. As both use an ECAPA-TDNN head, we also rescore search with an x-vector [Snyder et al., 2018], a GE2E d-vector [Wan et al., 2018] and WavLM-base+ [Chen et al., 2022] (App. E). ECAPA and the recognisers are used only for evaluation. Naturalness is UTMOS [Saeki et al., 2022]. An output is *degenerate* if synthesis failed or its transcript is empty, has under 30% of the reference characters, or repeats one 4-gram six or more times in a row. Each metric is averaged over all items, degenerate ones included. The *WER floor* is Whisper’s WER on the real target recordings (0.0345). The *SIM-o ceiling* is the SIM-o of the real target after a Mimi round trip (0.5554). Without the round trip, the real target scores SIM-o 0.678; the difference is what the codec loses. Both reference points are properties of our setup (real recordings under Whisper, and the Mimi codec), not hard limits: F5-TTS v1 Base [Chen et al., 2025b], which does not use Mimi, goes past both. At NFE 32 on our items it reaches WER 0.0248 (0.0860 under the second ASR), SIM-o 0.6549 and UTMOS 3.54. Because it differs from our models in data, size and output representation, we report it only as a reference for what a strong system scores on our items (a *calibration anchor*), not as a system to beat.

Data, design and statistics. Models are trained on a 2,000 h speaker-stratified subset of Emilia-EN [He et al., 2024] with 64,680 speakers. We evaluate cross-sentence zero-shot synthesis: prompt and target are different utterances of one speaker. The 400 test items come from 174 of 200 held-out speakers, with 4–15 s targets whose real recording Whisper transcribes at WER at most 0.25. The grid has 15 configurations: three parameter budgets with five *shapes* (width–depth splits at fixed N) each (App. C), trained at 3 seeds for 30k steps with μP [Yang et al., 2021] (effective batch 256, base learning rate 0.004). We vary $T \in \{1, 2, 4, 8, 16\}$ at evaluation only, which gives 75 points per metric from 45 models; $T=32$ and 64 were added later for every run (Table 1, App. G) and are not part of the registered fit. Let the error e be WER or $1 - \text{SIM-o}$, averaged over items and runs, and e^* the floor or one minus the ceiling. The *reachable range* runs from $e(1)$ to e^* , and refinement to T steps closes the share $s(T) = (e(1) - e(T)) / (e(1) - e^*)$ of it. The *ratio* is $s_{\text{WER}} / s_{\text{SIM}}$. All intervals are 95% percentile bootstrap intervals [Efron, 1979] over 2000 replicates. Unless a caption says otherwise, they resample runs, because the T points of a run share a model.

4 Results

4.1 Refinement closes more of the intelligibility range

Refinement from $T=1$ to the default $T=16$ closes 86.2% (CI [82.8, 89.1]) of the reachable intelligibility range and 46.4% (CI [45.0, 47.8]) of the reachable identity range (Table 1). The ratio of the two shares is $1.86 \times$ (CI [1.82, 1.90]). The ranges end at the measured WER floor and SIM-o ceiling (§3), and no model is fitted.

Table 1: *Share of the reachable range closed by refinement*, from $T=1$ to each T , over all 45 runs at 30k steps. Brackets: 95% intervals from a bootstrap over runs.

T	share of reachable range closed		
	intelligibility (WER)	identity (SIM-o)	ratio
2	32.2% [29.7, 34.4]	20.0% [19.1, 20.9]	$1.61 \times$
4	69.8% [66.0, 73.1]	35.7% [34.3, 37.0]	$1.96 \times$
8	81.8% [78.3, 84.8]	43.5% [42.1, 44.9]	$1.88 \times$
16	86.2% [82.8, 89.1]	46.4% [45.0, 47.8]	$1.86 \times$
32	87.9% [84.9, 90.5]	47.5% [46.2, 48.8]	$1.85 \times$
64	88.4% [85.3, 91.0]	47.9% [46.6, 49.2]	$1.85 \times$

The ratio is stable in T : from $T=4$ to $T=64$ it stays between $1.85 \times$ and $1.96 \times$. It is also stable under monotone maps of the error. With the original error and under $\log(1 + \text{err})$, $\sqrt{\text{err}}$ and err^2 it stays between $1.68 \times$ and $1.86 \times$ (Table 15). The registered fit’s main statistic, the difference $\Delta\tau$ between the two step exponents, is not: it changes sign under some of these maps (§4.5).

At $T=1$ almost every output is unintelligible: degenerate outputs have WER 1.124 and the others 1.148 (insertions push WER above one). Degenerate outputs make up 40.4% of outputs at $T=1$ and supply 39.0% of the WER drop to $T=16$. Dropping them therefore cannot show whether this poor start inflates the WER share; starting the count later can (App. D). From $T=2$, refinement closes 79.7% of the remaining intelligibility range and 33.0% of the identity range, a ratio of $2.41 \times$ (CI [2.33, 2.49]); from $T=4$ the ratio is $3.25 \times$. These are shares of smaller ranges, so only their ordering compares with Table 1. Keeping only items that are non-degenerate at every T (226 per run on average) gives $1.97 \times$ (CI [1.93, 2.01]). In each of the 6 variants of Table 7a above the F5-TTS row, at 30k and, where computed, at 90k steps, the intelligibility share is the larger one and the 95% interval on the ratio lies above one.

The ordering also holds with F5-TTS, the calibration anchor of §3, as the bound. With its WER as the floor and its SIM-o as the ceiling, the shares are 85.5% and 36.0%, a ratio of $2.37 \times$ (CI [2.32, 2.42]). F5-TTS does not emit Mimi tokens, so its SIM-o is not limited by our codec.

4.2 The ratio shrinks with training

Training longer shrinks the ratio, and the ordering holds at every length we tested (Table 2). We retrained 9 runs (C1, C3 and C5, 3 seeds each) to 90k and 180k steps. The ratio falls from $1.84 \times$ at 30k steps to $1.36 \times$ at 90k and $1.23 \times$ at 180k, and every interval lies above one. Longer training here means more passes over the same 2,000 h of speech, with no new data: 9.7, 29.1 and 58.2 passes.

Table 2: *The ratio shrinks with training.* The same 9 runs at three training lengths, $T=1$ to 16. Column 3 re-transcribes the same audio; intervals resample configurations and items (1000 replicates). The 180k row uses run records not yet in the public release (App. A).

training	intelligibility share		identity		
	Whisper	wav2vec 2.0	share	ratio	95% CI
30k	90.6%	88.5%	49.3%	$1.84 \times$	1.73, 1.97
90k	97.1%	99.2%	71.5%	$1.36 \times$	1.28, 1.44
180k	97.0%	99.1%	78.8%	$1.23 \times$	1.18, 1.29

The intelligibility share reaches a plateau across training; the identity share does not. From 90k to 180k steps the intelligibility share changes by -0.09 points under Whisper and -0.06 points under the second recogniser, a CTC model of another family. The identity share rises from 71.5% to 78.8%, so the ratio falls. Because the intelligibility share is already 97.1% at 90k steps, further training can only move the ratio toward one; the ordering at 180k says that refinement leaves more of the identity range open, not that the gap between the two shares stays large.

The ratio is also smaller at the largest budget. At 30k steps it is $1.91 \times$ (CI [1.80, 2.00]) at 20 M, $1.83 \times$ and $1.83 \times$ at 50 and 125 M, and $1.66 \times$ (CI [1.63, 1.69]) at 276 M, which has two seeds per shape (Table 7b). Prompt CFG also shrinks it (§5). We claim the ordering, which holds at every budget and training length we tested. Three training lengths do not show where the ratio converges, so we do not extrapolate it.

4.3 Search raises identity at matched NFE

At every matched-NFE tier we tested, best-of- K search gives higher speaker similarity than refinement, and refinement gives lower WER (Table 3). At NFE 128, best-of-2 search beats the default $T=16$ by +0.0153 in ECAPA similarity (CI [0.0127, 0.0179]) and wins on 57.4% of items, with WavLM-L as the selector. With the roles swapped, so that ECAPA selects and WavLM-L scores, search gains +0.0089 in SIM-o (CI [0.0065, 0.0114]) and wins on 53.9% of items. Selecting and scoring with the same encoder would be circular, so we always use two different ones.

Both pairings of selector and scorer beat refinement at every NFE we tested, and search wins on more than half of the items in each case (95% intervals in App. E); the gains are smaller when ECAPA selects. The two selectors pick the same candidate on only 64.3%, 43.4% and 29.3% of items at $K = 2, 4, 8$ (chance: 50.0%, 25.0%, 12.5%). At NFE 512, the four speaker encoders other than the selector all prefer search, on 64.6–79.0% of items.

Table 3: Search against refinement at matched NFE (15 runs, 200 items). Each candidate is drawn at $T=8$ (64 NFE), so NFE = $64K$. WER is absolute, for the WavLM-L-selected output.

NFE	T vs K	WavLM-L selects, ECAPA scores		ECAPA selects, WavLM-L scores		WER refinement / search
		Δ similarity	95% CI	Δ SIM-o	95% CI	
128	16 vs 2	+0.0153	[0.0127, 0.0179]	+0.0089	[0.0065, 0.0114]	0.1530 / 0.1938
256	32 vs 4	+0.0322	[0.0307, 0.0338]	+0.0195	[0.0170, 0.0218]	0.1335 / 0.1947
512	64 vs 8	+0.0444	[0.0416, 0.0470]	+0.0294	[0.0264, 0.0325]	0.1316 / 0.1955

Table 4: SIM-o gain (WavLM-L) from each lever, against the headroom left between our best 90k-step model and the codec ceiling. Each row is one bar of Fig. 2A. Costs are in WER points (one point is 0.01 absolute WER).

lever	change	cost	Δ SIM-o
refinement ^a	$T=1 \rightarrow 16$	$16 \times$ NFE	+0.1606
training	30k $\rightarrow$ 90k steps	$3 \times$ training steps	+0.0827
parameters	best shape, 20 M $\rightarrow$ 276 M	larger model	+0.0531
search ^b	best-of-2 vs $T=16$, matched NFE	+4.08 ^c , +7.1–7.9% compute	+0.0089
search ^b	best-of-8 vs $T=64$, matched NFE	+6.39 ^c	+0.0294
guidance	$\gamma=0.5$ at $T=16$ vs $T=32$	+4.41	+0.0087
rate matching	length from the prompt’s rate	+0.86	+0.0044
headroom	best 90k-step model $\rightarrow$ codec ceiling		0.0747

^aFrom one-step decoding, not an operating point. ^bECAPA selects. ^cWER of the WavLM-L-selected output, not of the ECAPA-selected one whose gain is shown; the WER of the ECAPA-selected output was not stored.

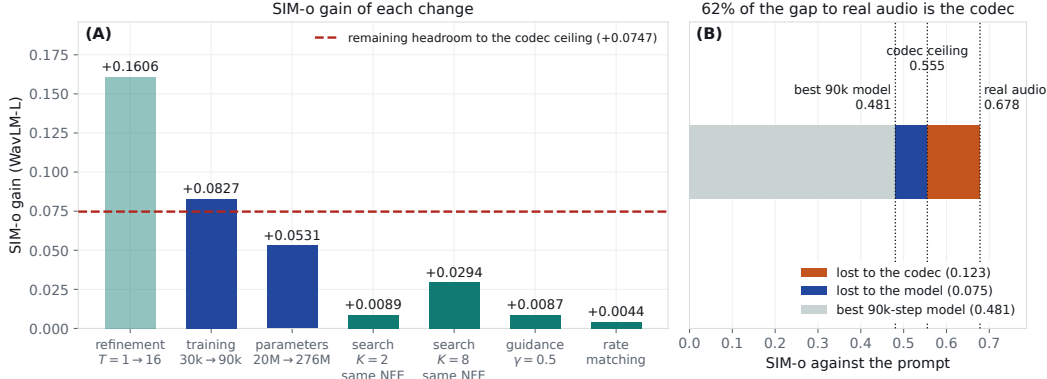


Figure 2: (A) SIM-o gains of Table 4; search bars are ECAPA-selected and at matched NFE, refinement (faded) starts at $T=1$. (B) 62% of the gap to real audio is lost in the Mimi codec.

Search has two costs. First, it raises WER: the best-of-2 output picked by WavLM-L has +0.0408 higher absolute WER than $T=16$ refinement. More candidates do not lower it; more refinement steps do (Table 3). Second, selection costs compute, because every candidate is decoded by Mimi and embedded by WavLM-L. At $K=2$ this adds 7.1–7.9% to the FLOPs of generation, as much as 9.1–10.2 generator passes. It is at most 17.4% if phoneme tokens are not counted, and the estimated CPU wall time is $1.09 \times$ that of refinement. ECAPA and the ASR are used only for evaluation and add nothing. Refinement given more total compute than search still loses: best-of-2 beats $T=32$ by +0.0205 (App. E).

The gap grows to +0.0322 at NFE 256 and +0.0444 at NFE 512, partly because refinement itself gets worse past $T=16$: its ECAPA similarity changes by -0.0052 from $T=16$ to $T=32$ and by -0.0079 to $T=64$. The NFE 512 result holds on the 276 M budget (+0.0422; +0.0284 with ECAPA selecting) and after $3 \times$ training (+0.0368; +0.0261). At NFE 128 after $3 \times$ training, search still gains +0.0116 with WavLM-L selecting, but the swapped control’s interval includes zero (+0.0085, CI $[-0.0006, 0.0237]$, 3 runs; App. E).

4.4 What else raises speaker similarity

Apart from refinement itself, whose gain is counted from unusable one-step output, training longer raises SIM-o most and adding parameters comes second; search and guidance gain much less (Table 4, Fig. 2). Retraining three configurations from 30k to 90k steps raises SIM-o by +0.0827 (CI [0.0733, 0.0908], 9 runs), and, for one run (C3, seed 0), doubling again to 180k adds +0.0155 (CI over items [0.0064, 0.0243]). Moving from the best 20 M shape to the best 276 M shape at 30k steps adds +0.0531.

Speaker-contrastive guidance runs a second forward pass per step, so guided $T=16$ costs NFE 256 and is compared with unguided $T=32$. At NFE 256, best-of-4 search with ECAPA selecting gains +0.0195 over unguided $T=32$, and guidance at $\gamma=0.5$ gains +0.0087 (CI [0.0063, 0.0115]). At matched NFE, search’s gain grows with K : +0.0089, +0.0195 and +0.0294 at $K = 2, 4, 8$ (Table 3). Raising γ does not help guidance: its gain is -0.0021 at $\gamma=1.0$ and -0.0322 at $\gamma=2.0$ (App. F). Guidance also costs +4.41 WER points, above the registered limit of 2.0 points, so its registered test (H-S4) fails (App. B).

Rate matching and longer prompts add little (App. F). Setting each item’s length from its own prompt’s speaking rate gains +0.0044. Prompts longer than the 3 s used in training do not raise SIM-o, and at 9 s they raise WER by 71.7 points. Finally, 62% of the distance between our best 90k-step model and real audio is lost in the Mimi codec (Fig. 2B).

4.5 What the registered fit identifies

Before training, we registered four hypotheses about a power-law fit of error over width w , depth d and refinement steps T (App. B), two of them declared primary (H-D2 and H-D3). H-D2 passes as declared; H-D3 and H-D4 fail their registered criteria; H-D1 cannot be decided, because the width exponent is not identified. App. G gives the two fitted forms, the exponents and the fit diagnostics. None of this affects the share in §4.1, which uses no fitted model.

Steps and the two metrics (H-D2). The first registered primary test passes: the step exponent is larger for WER than for SIM-o, by $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$ (CI [0.092, 0.131]). Its sign depends on how the error is scaled. When we refit the same measurements after a monotone change of the error, such as $\sqrt{\text{err}}$, $\Delta\tau$ turns negative in 3 of the reparameterisations in Table 15. It also depends on the range of T the fit covers. We report $\Delta\tau$ and do not rely on it.

Depth and steps (H-D3). This is the second registered primary hypothesis. We could not identify a fixed exchange rate between depth and steps. The registered substitution form assumes that error depends on depth and steps only through dT^κ , as if doubling T multiplied the effective depth by a fixed factor 2^κ . Its exponent κ sits at its upper bound in 14 of 14 fits and in all 2000 bootstrap replicates. This form fits worse than the separable one, by $\Delta\text{AICc} = +69.3$ for WER and +71.3 for SIM-o [Hurvich and Tsai, 1989]. The registered criterion was at most +4, so H-D3 fails. Neither form fits the data well. The weighted χ^2 per degree of freedom is 36.8 (WER) and 23.3 (SIM-o) for the substitution form, and 14.3 and 8.8 for the separable form. So the AICc gap compares two misspecified forms.

Shape at fixed parameter count (H-D1). Shape matters, but we could not identify how. At $T=16$, a fit with separate width and depth terms beats a fit in the parameter count N alone, with $\Delta\text{AICc} = -15.5$ for WER and -5.5 for SIM-o (registered threshold -4). The registered test compares the ratio $\rho = \alpha/\beta$ of the width and depth exponents across the two metrics. In all four registered fits, and in 22 of 33 fits overall, the coefficient of the width term reaches the upper limit set for it, so the data cannot pin down the width exponent α , and therefore not ρ either.

Depth optimum and extrapolation (H-D4). The WER-best depth is the deepest shape tested in 9 of the 15 (budget, T) cells, so in those cells the grid bounds the WER-optimal depth only from below. At $T=16$, no budget’s WER optimum is resolved: the largest margin, 3.6 WER points for the deepest shape of the 125 M budget, is under two standard errors of the difference between two configuration means (SE 2.3), and in the two smaller budgets the two best shapes are within one standard error of each other. Fits on the two smaller budgets also do not predict the largest (Fig. 5).

For WER the mean absolute percentage error is 20.2%, above the registered 15%. For SIM-o the fitted form does no better than the N -only model (2.4% against 2.2%).

5 Limitations

All results come from one small regime. The grid has 15 configurations at 3 seeds, with 19–133 M non-embedding parameters, plus one 276 M budget. The grid models are trained for 30k steps on 2,000 h of English speech from one corpus (Emilia-EN), with one codec (Mimi) and one confidence sampler. The 90k and 180k runs make more passes over the same data. The ratio shrinks with training and is smaller at the largest budget and under prompt CFG (§4.2), so we claim only the ordering, and only in this regime. A deeper grid is needed to locate the WER-optimal depth (§4.5).

All metrics are automatic: ASR word error, speaker-encoder similarity and UTMOS. We ran no listening test, so we cannot say whether a similarity gain of +0.0153 or +0.0089 is audible. We did not evaluate MaskGCT, a masked-diffusion TTS system, on our items. F5-TTS is a calibration anchor: it differs from our models in data, scale, codec and vocoder. Target length comes from a corpus-wide character rate; per-item rate matching gave a small mean gain that a per-item sign test did not confirm (Table 11). The selector (WavLM-L) and the ECAPA scorer both use an ECAPA-TDNN speaker head [Desplanques et al., 2020] trained for verification on VoxCeleb data [Nagrani et al., 2017, Chung et al., 2018], so they may share biases. Swapping their roles keeps search ahead of refinement at matched NFE (Table 3), but at NFE 128 after $3\times$ training the swapped interval includes zero (App. E). We did not try GE2E, x-vector or WavLM-base+ as selectors, because they scored only the candidate WavLM-L selected. The selection cost omits resampling and the speaker head of WavLM-L (App. E).

The main grid uses no classifier-free guidance (CFG) [Ho and Salimans, 2022]. A separate registered check retrained 9 runs with condition dropout and compared prompt CFG at $T=16$ with unguided $T=32$ at matched NFE. With guidance the ordering still holds, but the ratio falls from $1.87\times$ (the same runs without guidance) to $1.31\times$ (CI [1.28, 1.34]), so the factor depends on the sampler as well as on scale. Guidance raised SIM-o by +0.0483 (CI [0.0442, 0.0520]) at a cost of +0.0758 absolute WER (CI [0.0660, 0.0891]), more WER than the registered limit allows; we did not run search on these models.

We registered four hypotheses about the fitted surface, two of them primary (§4.5). The step-exponent test passed, but its sign depends on the error coordinate. The exchange-rate test failed its AICc criterion with κ at its bound, and the shape test could not be decided because the width coefficient sits at its cap. The extrapolation test failed. The share of the reachable range, our main measurement, was added later (protocol v1.4), after $\Delta\tau$ was found to depend on the coordinate. It is post hoc, not a registered test (App. B).

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A Reproducibility and release

Code. <https://github.com/nityanandmathur/aspect-d> contains the model, training and sampling code, the evaluation and fitting code, the run records and per-item scores of the released runs, the pre-registration documents, and scripts that rerun each stage. REPRODUCE.md names the script and the input files behind every table and figure. The code is released under the MIT licence and the model weights under CC BY-NC 4.0.

Models. The model repository holds the checkpoints of all 45 grid runs and of the 20 learning-rate sweep runs, each with its configuration and training record. It also holds extension checkpoints at the 276M budget and at 90k training steps, and the measured surface with its fits: <https://huggingface.co/nityanandmathur/aspect-d-masked-diffusion-tts>. It also holds audio samples at $T=1$ and $T=16$ with their prompts and real targets.

Compute. Training the 45 grid runs took 162.5 GPU-hours on NVIDIA B200 GPUs. With the learning-rate sweep (6.9) and the synthesis for the grid (2.4), the main study used 171.8 GPU-hours. The extension runs with released records add 169.3 GPU-hours of training. Later runs without GPU-hour records in the repository are not counted: most of the 180k-step runs, the classifier-free guidance check and the F5-TTS anchor. All GPU-hour values are sums of each job’s wall-clock time. Small jobs often shared one GPU, so the sums overstate exclusive GPU time.

Where the numbers come from. Measured values in the text, tables and figures are LaTeX macros that scripts in `src/` write from committed artifacts. On a CPU, `recipes/reproduce_paper_cpu.sh` regenerates the macro files from the released records and compares them with the copies used here. The run records behind the 180k-step row of Table 2 are not yet part of the public release, so the recipe reports those values as unverifiable. Design constants, such as step counts, T , K , widths and depths, are typed.

B Pre-registration ledger

Table 5 lists the hypotheses of the first three registrations and their verdicts. Each registration was committed before any data that test it existed: v1.0 (H-D) before the first training run, v1.1 (H-E) and v1.2 (H-S) before their extension runs. A verdict of DISCORDANT means the hypothesis’s two registered tests disagree in direction.

Table 5: Registered hypotheses of v1.0 to v1.2 and their verdicts.

id	claim	verdict
H-D1	$\Delta\rho > 0$: shape affects the two metrics differently at $T=16$	not supported [†]
H-D2*	$\Delta\tau > 0$: error falls faster in T for WER than for SIM-o	supported [§]
H-D3*	depth-step exchange rate κ	not supported [†]
H-D4	small budgets predict the largest	not supported
H-E1a	$T_{\text{WER}}^* > 16$	supported
H-E1b	extended τ inside the v1.0 CI	refuted
H-E2	iso-latency allocation is depth-first	refuted
H-E3	coarse-level NFE beats fine	supported
H-E4a	scale persistence, four budgets	supported
H-E4b	predicted vs observed within 15 %	refuted
H-E5	survives $3\times$ training compute	supported
H-S1	longer prompt context raises identity	discordant
H-S2	search raises identity	supported
H-S3	rate-matched length raises identity	discordant
H-S4	contrastive guidance raises identity within $+2$ WER points	refuted [‡]

*Declared primary in v1.0. §Passes as declared; its sign depends on the error coordinate (§4.5).

[†]Not identified: the parameter sits at a bound (§4.5); H-D3 also fails its AICc criterion.

[‡]SIM-o rises at $\gamma=0.5$, but WER rises by more than the registered 2.0 points (§4.4).

Some registered items have no row. H-S0 classifies the codec headroom rather than testing a claim; its value is the headroom in §4.4. H-E6, on confidence-adaptive stopping, was gated on the compute schedule and not run. Two v1.3 hypotheses are supported: search still beats refinement on the 276 M and 90k-step checkpoints (H-T1), and SIM-o still rises from 90k to 180k steps (H-T2). The third (H-T3) was withdrawn, because a code error meant the retrained model never saw the intended variable-length prompts. The v1.5 guidance check is reported in §5. The share of the reachable range (§4.1) is not a registered test: it was introduced at v1.4, after the registered analysis.

C The shape grid

Table 6 lists every configuration of the grid with its mean at $T=16$. Each budget holds five shapes with about the same non-embedding parameter count $N = 12dw^2$: 20, 50 and 125 M. Degenerate outputs follow the rule of §3.

Table 6: The shape grid, with means at $T=16$ over seeds 0, 1, 2 and 400 evaluation items per run.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	degenerate (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

D Robustness of the share of the reachable range

Table 7 recomputes the shares of Table 1 under the variants of §4.1. Each variant averages over items and runs at both ends and then takes the share. The rows that start from $T=2$ or $T=4$ give the share of the range from that T to the floor or ceiling that $T=16$ closes. These ranges are smaller than the range from $T=1$. The unpaired non-degenerate row averages each end over its own non-degenerate items. The paired row keeps, per run, the items that are non-degenerate at both $T=1$ and $T=16$. The next row keeps the items that are non-degenerate at every T from 1 to 16. The items column gives the mean number of items per run kept at both ends, and every non-degenerate row keeps at least 197 items in each run. The paired and every- T rows use the mean SIM-o ceiling of the items they keep. Per-item WER floors were not stored, so every row uses the floor over all 400 items, with two exceptions. The floor-zero row sets the floor to zero, which lowers the intelligibility share, so it is the cautious case. The F5-TTS row takes the WER and SIM-o of F5-TTS on the same items as floor and ceiling. Only 12 items are non-degenerate in every run at every T , too few to report a fixed item set.

Part (b) gives the shares and the rate of degenerate outputs by parameter budget; budget D has two seeds per shape. Longer training does not lower the rate at $T=1$: for the 9 runs of Table 2 it is 39.7% at 30k steps and 41.5% at 90k. At 90k it is 4.2% at $T=2$ and 0.1% at $T=16$.

Table 7: *The ordering holds in every variant.* (a) Shares closed by $T=16$, with 95% intervals on the ratio. Left: all 45 grid runs, with the bootstrap of Table 1. Right: the 9 runs of Table 2, with its bootstrap. (b) By parameter budget at 30k steps, from $T=1$.

(a) Variants						
variant	all 45 runs, 30k steps				9 runs: ratio	
	items	WER share	SIM-o share	ratio	30k steps	90k steps
from $T=1$ (Table 1)	400	86.2%	46.4%	$1.86 \times [1.82, 1.90]$	$1.84 \times [1.73, 1.97]$	$1.36 \times [1.28, 1.44]$
from $T=2$	400	79.7%	33.0%	$2.41 \times [2.33, 2.49]$	$2.45 \times [2.23, 2.72]$	$1.70 \times [1.55, 1.87]$
from $T=4$	400	54.3%	16.7%	$3.25 \times [3.00, 3.53]$	$3.57 \times [2.99, 4.46]$	$2.92 \times [2.46, 3.52]$
non-degenerate, unpaired	238	86.5%	45.9%	$1.88 \times [1.84, 1.93]$	—	—
non-degenerate, paired	238	87.5%	44.6%	$1.96 \times [1.92, 2.01]$	$1.92 \times [1.79, 2.08]$	$1.44 \times [1.35, 1.54]$
non-degenerate at every T	226	87.9%	44.5%	$1.97 \times [1.93, 2.01]$	$1.94 \times [1.80, 2.09]$	$1.44 \times [1.36, 1.54]$
same, WER floor at zero	226	85.2%	44.5%	$1.91 \times [1.87, 1.95]$	—	—
F5-TTS as the bounds	400	85.5%	36.0%	$2.37 \times [2.32, 2.42]$	—	—

(b) By parameter budget, 30k steps									
budget	WER share	SIM-o share	ratio	degenerate outputs (%) at $T =$					
				1	2	4	8	16	
A (20 M)	80.3%	42.0%	$1.91 \times [1.80, 2.00]$	40.0	10.7	1.5	0.4	0.2	
B (50 M)	88.6%	48.4%	$1.83 \times [1.80, 1.87]$	41.2	8.1	0.5	0.2	0.1	
C (125 M)	89.8%	49.1%	$1.83 \times [1.79, 1.87]$	39.9	7.4	0.5	0.1	0.1	
D (276 M)	92.5%	55.7%	$1.66 \times [1.63, 1.69]$	42.5	7.3	0.3	0.1	0.1	
A–C pooled	86.2%	46.4%	$1.86 \times [1.82, 1.90]$	40.4	8.8	0.8	0.2	0.1	

E Search: cost and selector controls

Unless stated otherwise, every search result uses the 15 runs of the 125 M budget and 200 evaluation items per run. Every candidate is drawn at $T=8$ (64 NFE), and candidates differ only in their sampling noise. Best-of- K search therefore uses $\text{NFE} = 64K$ and is matched to refinement at $T=8K$. A gain is a difference of per-run means, with a bootstrap over runs (2000 replicates). A win rate is the share of items on which search scores higher, with a bootstrap over items.

Selection cost. Refinement decodes its output once and runs no selector. Search decodes all K candidates with Mimi and embeds each of them, and the prompt once, with WavLM-L. We count the generator’s FLOPs analytically and check the count against PyTorch’s FLOP counter. We count Mimi and WavLM-L with the same counter on randomly initialised copies, on one item of median length, and scale to the mean item length. One forward pass costs 69.7–78.5 GFLOPs for the generator (range over the five configurations), 39.8 for a Mimi decode and 276.9 for WavLM-L.

The generator’s input includes the phoneme tokens, whose number we estimate at 0.83 phones per character (espeak-ng). Leaving them out makes the generator cheaper and gives the upper bound in Table 8. The count leaves out the ECAPA-TDNN-small head on WavLM-L and the 24-to-16 kHz resampling. The wall-time ratio sums single forward passes of each network, timed on one item on CPU for configuration C3; it is not a timed run of the whole sampler. The ECAPA scorer and the ASR are evaluation-only and are not part of any deployed cost.

Table 8: Selection cost of best-of- K search, over refinement at the same NFE. Ranges span the five 125 M configurations. The upper bound leaves the phoneme tokens out of the generator count. Wall time is relative to refinement.

K	NFE	extra generator passes	extra compute (%)	upper bound (%)	CPU wall time
2	128	9.1–10.2	7.1–7.9	17.4	1.09×
4	256	17.1–19.3	6.7–7.5	16.5	1.08×
8	512	33.3–37.5	6.5–7.3	16.1	1.07×

Equal total compute. We also compare search with the cheapest refinement arm whose total compute, counted as above, is at least that of search. For best-of-2 this arm is $T=32$, and search gains +0.0205 in ECAPA similarity (CI [0.0180, 0.0230]). For best-of-4 it is $T=64$, and search gains +0.0349 (CI [0.0322, 0.0373]). No refinement arm we ran costs as much as best-of-8. These gains are larger than the matched-NFE ones, because refinement’s ECAPA similarity falls past $T=16$ (its WavLM-L SIM-o rises slightly; Table 1). Refinement at $T=8$ and at $T=16$ differ by +0.0010 (CI [−0.0015, 0.0035]). So no refinement arm from $T=8$ to $T=64$ reaches the similarity of best-of-2.

Selector and scorer. Only WavLM-L and ECAPA scored all eight candidates of every item, so only they can act as selectors. Table 9 crosses the two. The default (WavLM-L selects, ECAPA scores) and the swapped control (ECAPA selects, WavLM-L scores) both beat refinement at every tier, and the win-rate intervals of both lie above one half. The swapped gains are smaller: +0.0089, +0.0195 and +0.0294, against +0.0153, +0.0322 and +0.0444. Rows in which one encoder both selects and scores are circular; we show them for comparison and draw no conclusion from them. The other three encoders of Table 10 (x-vector, GE2E and WavLM-base+) scored only the WavLM-L-selected output, so we could not test them as selectors.

Table 9: Search against matched-NFE refinement, for each selector and scorer (125 M budget). Each cell gives the gain, the win rate in parentheses and the 95% CI of the gain and of the win rate. The first two rows repeat Table 3 and add win rates. Last row: share of items on which the two selectors pick the same candidate.

selector → scorer	NFE 128	NFE 256	NFE 512
	$K=2$ vs $T=16$	$K=4$ vs $T=32$	$K=8$ vs $T=64$
WavLM-L → ECAPA	+0.0153 (57.4%)	+0.0322 (65.3%)	+0.0444 (72.3%)
default	[0.0127, 0.0179]; [55.6, 59.2]	[0.0307, 0.0338]; [63.7, 67.0]	[0.0416, 0.0470]; [70.6, 73.8]
ECAPA → WavLM-L	+0.0089 (53.9%)	+0.0195 (58.8%)	+0.0294 (62.7%)
swapped	[0.0065, 0.0114]; [52.1, 55.6]	[0.0170, 0.0218]; [56.9, 60.6]	[0.0264, 0.0325]; [61.1, 64.5]
WavLM-L → WavLM-L	+0.0292 (63.4%)	+0.0560 (75.2%)	+0.0803 (84.9%)
circular	[0.0264, 0.0322]; [61.7, 65.2]	[0.0535, 0.0583]; [73.6, 76.7]	[0.0778, 0.0828]; [83.6, 86.1]
ECAPA → ECAPA	+0.0326 (67.4%)	+0.0629 (82.0%)	+0.0861 (91.3%)
circular	[0.0296, 0.0352]; [65.8, 69.1]	[0.0604, 0.0655]; [80.6, 83.4]	[0.0834, 0.0886]; [90.2, 92.3]
same pick (chance 1/ K)	64.3% (50.0%)	43.4% (25.0%)	29.3% (12.5%)

Replications. We repeated the NFE 512 comparison on the 276 M budget (five configurations, two seeds each) and on three configurations retrained for 90k steps ($3\times$ the schedule, one seed each). With WavLM-L selecting and ECAPA scoring, search gains +0.0422 (CI [0.0393, 0.0448], wins on 70.7% of items) on the 276 M budget and +0.0368 (CI [0.0321, 0.0428], 68.7%) after retraining. With ECAPA selecting and WavLM-L scoring, it gains +0.0284 (CI [0.0238, 0.0334], 61.7%) and +0.0261 (CI [0.0218, 0.0284], 62.2%). At NFE 128 the retrained runs gain +0.0116 (CI [0.0043, 0.0227]) with WavLM-L selecting and +0.0085 (CI [−0.0006, 0.0237]) with ECAPA

selecting. With 3 runs the second interval includes zero, so at this tier the retrained runs do not confirm the swapped control.

Other speaker encoders. Table 10 rescores the NFE 512 comparison (best-of-8 against $T=64$, WavLM-L selecting) with five speaker encoders. Their cosines sit on different scales, so we compare win rates: the share of items on which an encoder prefers the search output. A win rate does not change under a monotone rescaling of an encoder’s cosine. All four encoders other than the selector prefer the search output on 64.6–79.0% of items, and every interval lies above one half. We did not repeat this rescoring at NFE 128 or 256.

Table 10: Search against refinement at NFE 512 (15 runs, 200 items each), rescored by ECAPA [Desplanques et al., 2020, Ravanelli et al., 2021], x-vector [Snyder et al., 2018] (both trained on VoxCeleb [Nagrani et al., 2017, Chung et al., 2018]), GE2E [Wan et al., 2018], WavLM-base+ and WavLM-L [Chen et al., 2022, Anastassiou et al., 2024].

encoder	family / role	win rate	95% CI	raw Δ
ECAPA	ECAPA-TDNN	72.3%	[70.6, 73.8]	+0.0444
x-vector	TDNN	79.0%	[77.5, 80.5]	+0.0075
GE2E d-vector	LSTM d-vector	69.1%	[67.5, 70.8]	+0.0164
WavLM-base+	WavLM, other scale	64.6%	[62.9, 66.3]	+0.0173
WavLM-L*	WavLM / <i>selector</i>	84.9%	[83.6, 86.1]	+0.0803

*selects the candidates, so this row is circular and is not counted as agreement.

F Other inference-time interventions

All arms in this appendix use the 15 runs of the 125 M budget and change one thing at inference: the sampler’s logits, the target length, the prompt length, or the split of refinement passes across codebook levels.

Guidance and rate matching. Speaker-contrastive guidance is training-free. At each step it computes a second set of logits with another item’s prompt and pushes the output away from them with weight γ . That second forward pass doubles the NFE, so guided $T=16$ is compared with unguided $T=32$, at 256 NFE each. Rate matching sets each item’s target length from the speaking rate of its own prompt, in place of the corpus rate. Table 11 gives both. Guidance raises SIM-o and costs WER. Rate matching gives a small mean gain that is not reliable item by item.

Table 11: Speaker-contrastive guidance and rate matching. Δ is against unguided $T=32$ (guidance, matched NFE) or against the corpus rate (rate matching); SIM-o is scored by WavLM-L.

intervention	setting	Δ SIM-o	95% CI	Δ WER (points)
guidance	$\gamma = 0.5$	+0.0087	[0.0063, 0.0115]	+4.41
	$\gamma = 1.0$	−0.0021	[−0.0059, 0.0023]	+8.94
	$\gamma = 2.0$	−0.0322	[−0.0367, −0.0274]	+19.10
rate matching	per-item rate	+0.0044	[0.0017, 0.0067]	+0.86
13 of 15 runs gain, but a per-item sign test gives $p = 0.051$.				

Prompt length. Training used 3 s prompts only. Table 12 sweeps the prompt length at inference, and every arm is scored against the same 3 s reference. No length raises SIM-o above the 3 s arm. At 6 and 9 s, SIM-o stays close to the 3 s arm while absolute WER rises to 0.7692 and 0.8642: the output keeps the speaker but no longer follows the text.

Allocating refinement passes across codebook levels. Table 13 holds the total at 32 NFE and moves the passes between the eight codebook levels. Coarse spends 25 passes on level 1 and one on each other level, fine does the reverse, and uniform gives every level four. The same 400 items are synthesised by every schedule, and the bootstrap (2000 replicates) resamples items. Coarse minus fine is -0.9289 in absolute WER (CI $[-0.9595, -0.8999]$). Against uniform, coarse has lower WER but also lower SIM-o and UTMOS, so the split of passes trades intelligibility against identity.

Table 12: Prompt length at inference (15 runs, 397 items). Every arm is scored against the same 3 s reference; WER is absolute.

prompt	SIM-o	WER
1.5 s	0.3213	0.1578
3 s (<i>training length</i>)	0.3839	0.1474
6 s	0.3724	0.7692
9 s	0.3759	0.8642

Table 13: Three per-level step schedules at a matched total of 32 NFE (15 runs, 400 items). WER is absolute; fine has $5.2\times$ the WER of coarse.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, . . . , 1]	0.2191	0.3057	2.17	0.1%
uniform [4, . . . , 4]	0.3144	0.3477	2.51	0.5%
fine [1, . . . , 1, 25]	1.1480	0.2246	1.56	39.1%

G Fitted surface

The registered analysis fits power laws to the error of each metric, with $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$. Part A uses the 15 configurations at $T=16$. It compares a fit with separate width and depth terms, M_{full} : $\text{err} = E + A w^{-\alpha} + B d^{-\beta}$, with a fit in the parameter count alone, M_N : $\text{err} = E + C N^{-\gamma}$. Part B uses all 75 surface points and compares the two forms of Eq. (1). In the separable form, steps add their own power law. In the substitution form, steps enter only through the product $d T^\kappa$.

$$\begin{aligned} \text{separable } (M_{\text{sep}}): \quad \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \\ \text{substitution } (M_{\text{sub}}): \quad \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m (d T^\kappa)^{-\beta_m} \end{aligned} \quad (1)$$

Table 14 gives the fitted exponents. Figure 3 shows the measured step curves behind the surface, and Figure 4 the fitted contours.

Table 14: Fitted exponents of Part B (M_{sep} for α, β, τ ; M_{sub} for κ), with run-level bootstrap intervals on τ . The last two columns compare forms by ΔAICc [Hurvich and Tsai, 1989].

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AICc}_{\text{sub-sep}}$	$\Delta\text{AICc}_{\text{full-N}}$
WER	1.574 [‡]	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919 [‡]	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at its bound. [‡]not identified: the width amplitude is at its cap.

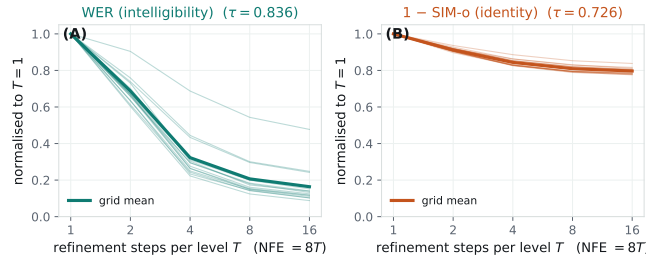


Figure 3: Error against refinement steps for all 15 configurations, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER falls by 0.952 in absolute terms from $T=1$ to 16. (B) $1-\text{SIM-o}$ falls by 0.161 over the same range. Same models, sampler and items; only T differs.

Coordinate and range. The step exponents change when the error is reparameterised (Table 15). $\Delta\tau$ is negative under 3 of the maps we tried. It also depends on the window of T the fit covers: +0.2191 over $T \leq 8$, +0.1102 over the declared $T \leq 16$, and +0.0603 and +0.0660 over $T \leq 32$ and

$T \leq 64$. The declared interval holds the fit’s $1/\text{SE}^2$ weights fixed. Resampling the weights with the runs widens it to $[0.0349, 0.1640]$.

Table 15: The same 75 measurements refitted after a monotone change of both errors. The last column is the share ratio of Table 1; — marks rows where it was not computed.

error coordinate	τ_{WER}	τ_{SIM}	$\Delta\tau$	share ratio
err (<i>pre-registered</i>)	0.8362	0.7260	+0.1102	$1.86\times$
$\log(1 + \text{err})$	0.6665	0.6913	−0.0248	$1.85\times$
$\sqrt{\text{err}}$	0.4325	0.6842	−0.2517	$1.68\times$
$\min(\text{err}, 1)$	0.7093	0.7260	−0.0167	—
err^2	1.5613	0.8122	+0.7491	$1.82\times$
$3 + 2 \text{ err}$ (v1.0 check)	0.7600	0.7249	+0.0351	—

Other refits of $\Delta\tau$. $\Delta\tau$ stays positive when one factor of the fit changes (Table 16): a fourth parameter budget, $3\times$ training compute, a different ASR, or amplitudes freed on a log scale. These rows test the sign of $\Delta\tau$ only. The log-amplitude refit does not identify α (see below). The released `fits.json` also holds an unweighted refit, which gives a positive $\Delta\tau$.

Table 16: $\Delta\tau$ when one factor of the fit changes. Each row refits the surface with that factor changed.

variation	what changes	$\Delta\tau$	95% CI
none (75 points, 45 runs)	—	0.110	[0.092, 0.131]
fourth budget, 276 M	10 further runs	+0.1095	[0.0951, 0.1241]
$3\times$ training compute	3 configs at 90k steps	+0.1767	[0.0789, 0.2854]
ASR swapped	Whisper-medium.en [Radford et al., 2023]	+0.1283	[0.1064, 0.1492]
amplitudes on a log scale	$\log A, \log B, \log C$ free	+0.1260	[0.1043, 0.1474]

Fit diagnostics. Each form is fitted by weighted non-linear least squares from 32 starting points (`src/fit.py`), and the bootstrap refits it on each resample of runs. Amplitudes are bounded to $[0, 10]$, exponents to at most 3, κ to $[-3, 3]$ and E to $[0, 1]$. The width amplitude A is at its cap in all four declared fits: 10.000 and 10.000 in Part A, and 9.999 and 10.000 in Part B, for WER and SIM-o. Across all committed fits it is at the cap in 22 of 33 and at zero in 2. In the bootstrap, A is at the cap in 1656 (Part A) and 1578 (Part B) of 2000 WER replicates, and in every SIM-o replicate. So only the product $A w^{-\alpha}$ is constrained, and α is not identified. Freeing the amplitudes on a log scale does not help: α_{WER} then sits at its upper bound (3.00, CI [2.87, 3.00]). κ is at its upper bound in 14 of 14 committed substitution fits, and in 2000 (WER) and 2000 (SIM-o) of 2000 replicates. E reaches zero for WER under M_{sep} . The depth amplitude B is interior in the declared fits, so β is identified: $\beta_{\text{WER}} = 2.040$ and $\beta_{\text{SIM}} = 0.553$. Points are weighted by $1/\text{SE}^2$, with SE taken over seeds and floored at a quarter of the pooled seed standard deviation; `fits.json` records how many points hit the floor. In all three budgets, the spread of configuration means at $T=16$ passes the registered power criterion for both metrics.

Depth optima. Table 17 gives the best depth per budget at $T=16$. For WER it is the deepest shape in budgets B and C, so those two values are lower bounds on the WER-optimal depth. No margin exceeds two standard errors of the difference between two configuration means. Over all five values of T in the grid, the WER-best depth is the deepest shape in 9 of the 15 (budget, T) cells of budgets A to C, and the SIM-o-best depth in 1.

Extrapolation. Figure 5 shows the registered extrapolation test (H-D4). For WER, the mean absolute percentage error is 20.2% for M_{full} and 20.4% for M_N , both above the registered 15% limit. For SIM-o it is 2.4% against 2.2%, so the fitted form does no better than the N -only model.

Serving latency. At batch 1, per-layer latency is flat in width on our device: 0.493–0.514 ms per layer for w from 256 to 1152. Serial latency is then $8 T d c_{\text{layer}}$, so at fixed parameter count a latency budget constrains only the product $T d$. The registered rule for splitting that product, depth before steps (H-E2), had to agree with the measured best split at 80% or more of the tested latency budgets,

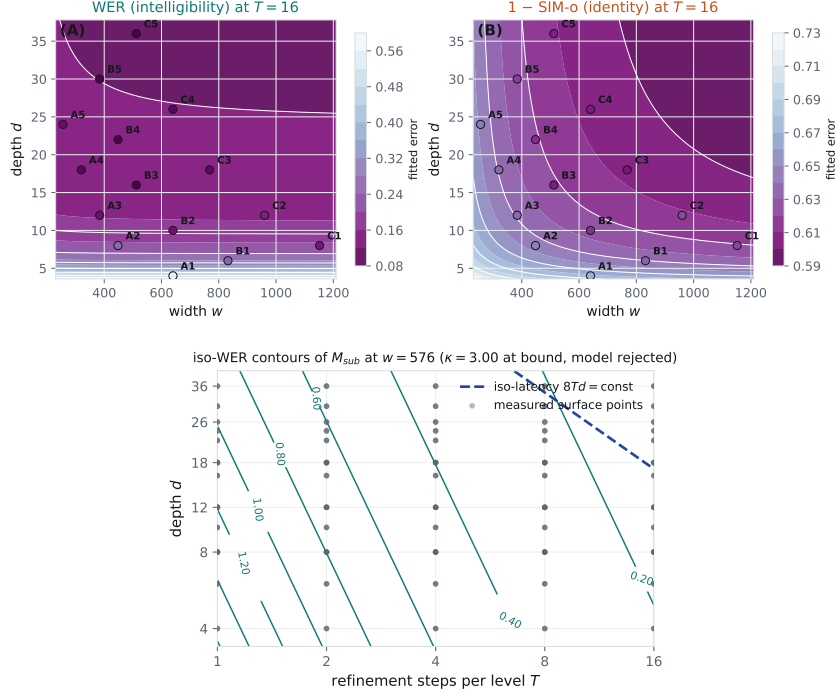


Figure 4: *Top*: fitted error contours of M_{full} in (w, d) at $T=16$, with the 15 configurations. The width amplitude is at its cap, so the contours' width direction is not identified. *Bottom*: iso-WER contours of M_{sub} at $w=576$ over the measured (T, d) points, with κ at its bound; the dashed line is one iso-latency curve ($8Td$ constant).

Table 17: Best depth per budget at $T=16$, from configuration means. Margin: WER points between the best and the runner-up shape; SE: standard error of a configuration mean; SE_{Δ} : of the difference of two means ($\sqrt{2}$ SE). Where the WER-best d equals the deepest d , it is a lower bound on the optimum.

budget	deepest d	WER-best d	margin	SE	SE_{Δ}	SIM-o-best d
A	24	18	1.7	4.3	6.1	12
B	30	30	0.3	1.5	2.1	16
C	36	36	3.6	1.6	2.3	18
D (276 M)	38	14	0.1	0.8	1.1	38

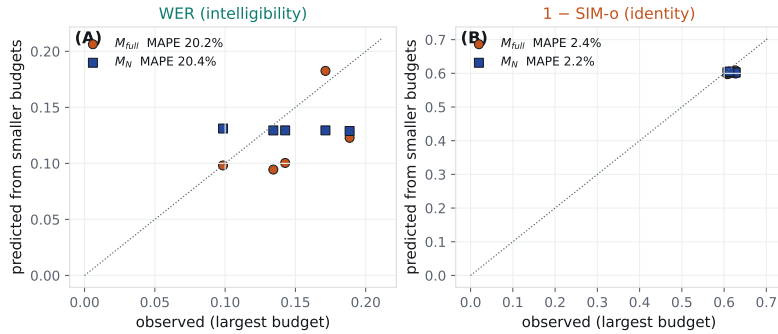


Figure 5: Extrapolation test (H-D4): M_{full} and M_N fitted on the two smaller budgets at $T=16$ and used to predict the largest. The dotted line marks a perfect prediction.

and it did not. The latency analysis stops at $T=16$, so we do not read this failure as evidence for the reverse rule.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness, for language refinement, and to redraft the camera-ready text from the submitted version and the reviews. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.